Class 3c: Review of concepts in Probability and Statistics

Business Forecasting

Hypothesis Testing

Hypothesis Testing

Let's go back to our question: can we get a higher price if cleanliness level is above 4.5?

This type of question can be answered with hypothesis testing, where we see which of two competing hypotheses is more likely given the data.

- Hypothesis is a claim about a parameter or a distribution.

1. H_0 Null Hypothesis:
    - Claim to be tested, the skeptical perspective
        - Prices of clean and dirty apartments are equal
        - $H_0 : \mu_c = \mu_d$
2. H_A Alternative Hypothesis:
    - An alternative claim under consideration
        - Prices of clean apartments are higher than prices of dirty apartments
        - $H_A : \mu_c > \mu_d$

- Assume the null is true and see if there is enough data to reject it in favor of the alternative
- You either reject the null, or fail to reject it (but never "accept" it)

Hypothesis Testing

Some examples of hypotheses testing:

- You released a new promotion to some customers (10% off) and you want to test if people who got the promotion spend different amount of money than those who didn't
  - H_0: People who got the promotion spend the same as people who didn't get the promotion
    - $H_0 : \mu_p = \mu_n$
  - H_A: People who got the promotion spend different amount than people who didn't get the promotion
    - $H_A : \mu_p \neq \mu_n$
- You designed a new healthy snack and want to test whether it needs "excessive sugar" sticker. It gets the sticker if it exceeds 100g of sugar. You take a sample and test.
  - H_0: The average sugar content is 100 or less
    - $H_0 : \mu_s = 100$
  - H_A: The average sugar content is more than 100
    - $H_A : \mu_s > 100$

Hypothesis Testing

- Suppose we calculated the sample mean for sugar content. It's either below or above 100g
- Why would we need a fancy test?
- Because we only have a sample, but we want to learn about the population parameter!
- Maybe in our sample it's below 100g, but in the whole population it's above
- Hypotheses are always about parameters! Never about sample statistics

Testing Procedure

A test procedure is a rule based on sample data for deciding whether to reject the null or not.

It relies on two elements:

- Test statistic
  - A function of the sample data that summarizes evidence for or against the hypothesis
  - We know how it is distributed under the null (e.g. standardized mean)
- Rejection region
  - Values of the test statistic for which we reject the null
  - These are values that are very unlikely under the null hypothesis
  - Typically threshold-based: reject H_0 if the test statistic > k

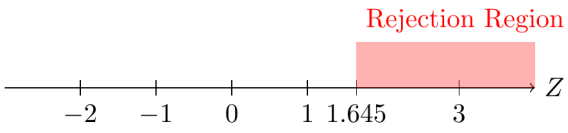
Testing Procedure: Example

Let's go back to the sugar example. Suppose we have a sample $n = 36$.

- Test statistic
  - Our test statistic could be:

$$z = \frac{\bar{x} - 100}{\underbrace{s_x/\sqrt{36}}_{SE}}$$

- Rejection region
  - For a 5% significance level, our rejection region would be:
  - $\{1.645, \infty\}$



Testing Procedure: Example

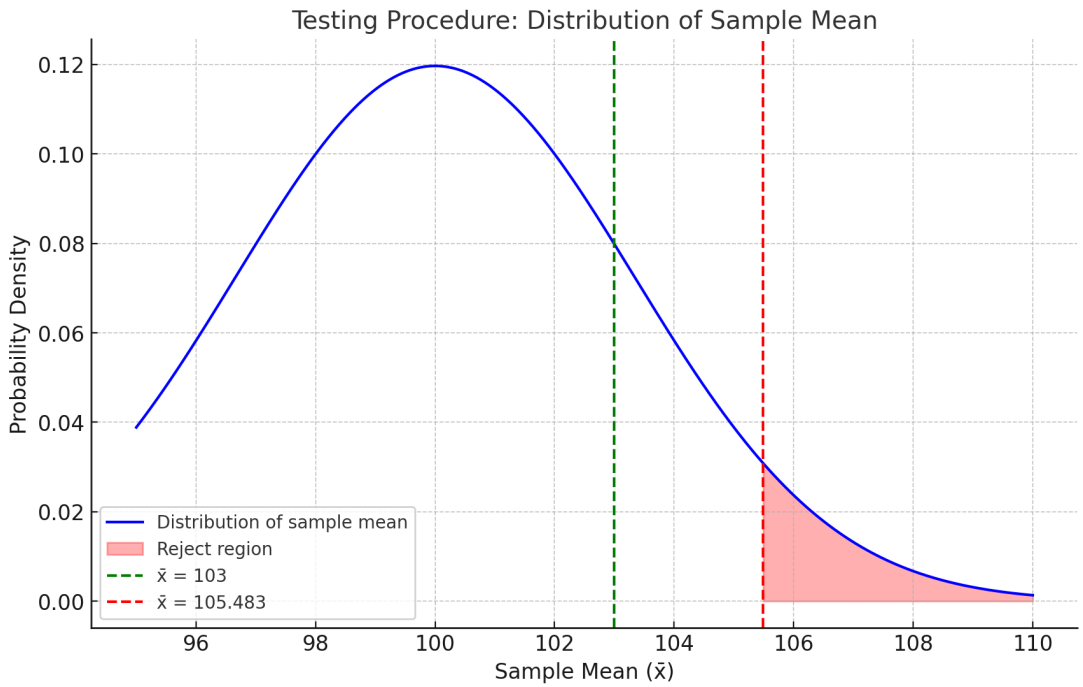
- Suppose I reject if $z > 1.645$
- Suppose $\sigma_x = 20$. How large does the mean need to be to reject the null?

Reject if:

- $\frac{\bar{x}-100}{20/\sqrt{36}} > 1.645$
- Or equivalently if $\bar{x} > 105.483$
- Those two formulations are equivalent
- Why don't we reject even if $\bar{x} = 103$?

Testing Procedure: Example

- Because there is randomness in the sample:
- Even if $\mu_s = 100$, we could get a sample with $\bar{x} > 100$



- Why $\bar{x} > 105.483$ though?
- Formally about it later, but it's very unlikely to get a sample mean that large if $\mu_s = 100$

Testing Procedure Summary

1. Set up your hypotheses
2. Calculate your test statistic
3. Check it against the rejection region
4. Decision: Reject or fail to reject the null hypothesis

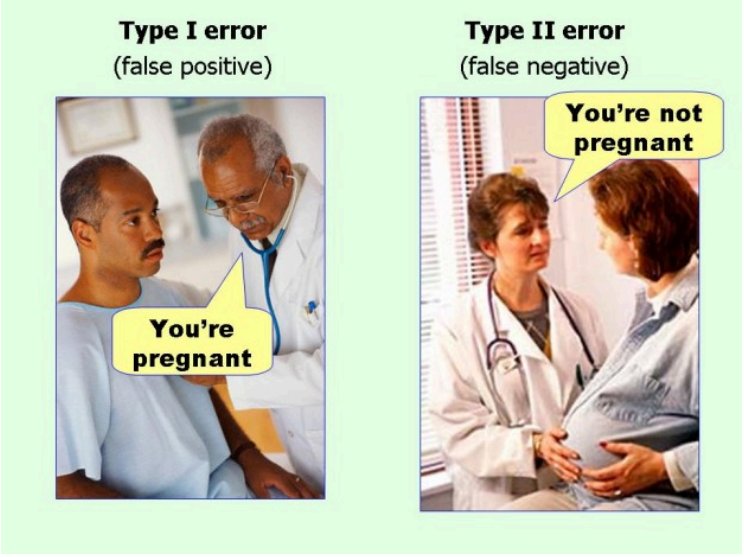
We will now go into the details of each step.

Decision errors

Decision errors

	Do not reject H_0	Reject H_0 in favor of H_A
H_0 true	Okay	Type 1 Error
H_A true	Type 2 Error	Okay

- H_0: You are not pregnant
- H_1: You are pregnant



Decision errors

Do people spend more if they get free shipping? (H_0 is that they spend the same)

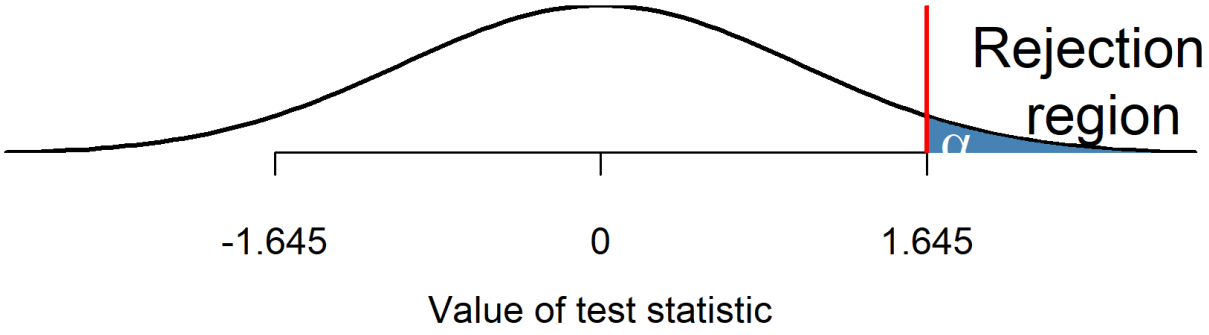
- Type 1 error: reject null when H_0 is true
  - Conclude that people spend more when they got free shipping, while in reality they spend the same as those who didn't
- Type 2 error: don't reject null when H_A is true
  - Conclude that people who got free shipping spend the same as those without free shipping, while those with free shipping spend more
- Imagine a trial of a murderer and suppose H_0 is innocent, and H_1 is guilty.
- Describe error type 1 and 2 in these scenarios
  - Type 1: Convict an innocent
  - Type 2: Do not convict a murderer
- Probability of making type 1 error is α
  - It's also called size, or significance level of the test
  - We don't want to incorrectly reject null more than α*100 percent of times.
- Probability of making type 2 error is β
  - To calculate β, you need to know the true μ

Probability of type 1 error

- In the sugar sticker example, suppose that the null is true and $\mu_s = 100$
- You take a sample of 36.
- Example: You were very unlucky in your sampling, by chance you got a sample with those rare packages with a lot of sugar and your $\bar{x} = 106$ which means $z = \frac{106-100}{\frac{20}{\sqrt{36}}} = 1.8$. According to the test procedure, you reject the null. You make an error.
- Q: What is the probability that the (standardized) mean of these 36 is in the rejection region (larger than 1.645)?
- What's the probability of making such error?

$$\alpha = P_{H_0}(\bar{X} > 105.483) = P_{H_0}\left(\frac{\bar{X} - 100}{20/\sqrt{36}} > \frac{105.483 - 100}{20/\sqrt{36}}\right)$$

$$\overset{\text{CLT}}{=} P(Z > 1.645) = 1 - \Phi(1.645)$$



- Higher critical value = lower chance of mistake (lower α)
- Lower critical value = higher chance of mistake (higher α)
- Choosing critical value determines α
- Or symmetrically, choosing α determines critical value
- We usually choose α and this will determine the rejection region.
- What would be the critical value for $\alpha = 0.1$?
- What would be the critical value for $\alpha = 0.01$?
- What would be the critical value for $\alpha = 0$?

LearnR: Critical values — use `qnorm()` to find the critical value for a given `(\alpha)`

Probability of type 2 error

- What's the probability our sample shows mean sugar content below critical value, if the true sugar content is above 100?
- What is the probability we do not reject the null, although the alternative is true?
- If I take 10000 samples from distribution with true mean > 100, what share of sample means will fall below the rejection region?
- In our example

$$\beta = P(\text{Type 2 error}) = \underbrace{P_{H_A}(z < 1.645)}_{\text{Pr. if alt is true}} = P_{H_A}(\frac{\bar{x} - 100}{20/\sqrt{36}} < 1.645)$$

  - Problem: How is $\frac{x-100}{20/\sqrt{36}}$ distributed if the alternative is true?

- CLT does not apply to: $\frac{\bar{x}-100}{20/\sqrt{36}}$ because 100 is not the true mean under the alternative

- To calculate it, we need to specify what is the true mean. Say $\mu_s = 110$

- Then CLT applies to: $\frac{\bar{x}-110}{20/\sqrt{36}}$

Probability of type 2 error

- We need to rewrite it so we have $\frac{\bar{x}-110}{20/\sqrt{3}6}$ on the left-hand side.

$$\beta = P_{H_A}(\frac{\bar{x}-100}{20/\sqrt{3}6} < 1.645) = P_{H_A}(\bar{X} < 105.483) = P\left(\frac{\bar{X}-110}{20/\sqrt{36}} < \frac{105.483-110}{20/\sqrt{36}}\right)$$

- Now this becomes:

$$P\left(z < \frac{105.483-110}{20/\sqrt{36}}\right) = P(z < -1.3551)$$

- Where z is normally distributed thanks to CLT

Probability of type 2 error

- More generally for this type of test:

$$\beta = P_{H_A}(\bar{X} < \mu_0 + z_\alpha \tfrac{\sigma_x}{\sqrt{n}}) = P\left(\frac{\bar{X} - \mu_s}{\sigma_x/\sqrt{n}} < \frac{\mu_0 - \mu_s}{\sigma_x/\sqrt{n}} + z_\alpha\right)$$

- Depends on what true μ is and where our rejection region is!
- When you decrease probability of type 1 error, you increase the probability of type 2 error

> LearnR: Type 2 error — simulate samples when the null is false and count how often you fail to reject

Trade-off between type 1 and 2 errors

$$\beta = P(\text{Type 2 error}) = P(Z < \frac{\mu_0 - \mu_s}{\sigma_x/\sqrt{n}} + z_\alpha) = \underbrace{\Phi(\frac{\mu_0 - \mu_s}{\sigma_x/\sqrt{n}} + z_\alpha)}_{\text{Normal CDF}}$$

$$\alpha = P(\text{Type 1 error}) = P(Z > z_\alpha) = 1 - \underbrace{\Phi(z_\alpha)}_{\text{Normal CDF}}$$

If I change critical value z_α, it will have the opposite effect on them.

Interactive visualization: https://shiny.rit.albany.edu/stat/betaprob/

Power of a test

- Power of a test is the probability of rejecting if alternative is true
- Power will be different for any possible value of the alternative and hypothesis
- Power for one sided test

$$Power = P_{H_A}(Reject) = 1 - \underbrace{P_{H_A}(\text{Not Reject})}_{\text{type 2 error}} = 1 - \beta$$

$$Power = 1 - P(z < \frac{\mu_0 - \mu_s}{\sigma_x / \sqrt{n}} + z_\alpha)$$

Interactive visualization: https://shiny.rit.albany.edu/stat/betaprob/

Power increases if:

- n increases
- Alternative is further away from the null
- σ decreases
- α increases

A/B Test Power: Professional Photos

- Airbnb wants to test if professional photos increase weekly revenue
- Design:
  - Control: no photos (baseline $\mu_0 = 2000$)
  - Treatment: professional photos (suspected $\mu_a = 2200$)
  - $\sigma = 600$, $n = 100$ per group, $\alpha = 0.01$
- Steps:
  1. State hypotheses: $H_0 : \mu_T - \mu_C = 0$ vs $H_A : \mu_T - \mu_C > 0$
  2. Find the critical value
  3. Compute power $= 1 - \beta$
  4. Find minimum n for 80% power

> LearnR: Power exercise — Design the Airbnb photography A/B test, calculate power, and find minimum sample size

General Procedure

1. Determine the null and alternative hypotheses
2. Collect your sample of independent observations
3. Choose the appropriate test
4. Calculate the test statistic
5. Compare it to the rejection region
6. Reject or fail to reject the null hypothesis

Let's talk about appropriate tests!

Choosing the Appropriate Test

The choice of test statistic and rejection region depends on:

- Type of data we are testing?
  - Single sample
    - Example of hypothesis: $H_0 : \mu = \mu_0$ vs $H_A : \mu \neq \mu_0$
  - Two independent samples
    - Example of hypothesis: $H_0 : \mu_{s1} = \mu_{s2}$ vs $H_A : \mu_{s1} \neq \mu_{s2}$
  - Paired data
    - Example of hypothesis: $H_0 : \mu_{d1} = \mu_{d2}$ vs $H_A : \mu_{d1} \neq \mu_{d2}$
- Can we estimate standard deviation well?
  - Yes
    - Large sample
    - Known standard deviation
  - No
    - Small sample but normal distribution

Hypothesis Testing: Single Sample

Single Sample Test for Mean

General idea:

- We test if the parameter is equal/larger/smaller than some concrete value
  - Ex: $H_0 : \mu = 3$ vs $H_A : \mu \neq 3$
- Test statistic: normalized sample mean

$$\text{test statistic} = \frac{\bar{X} - \mu_0}{SE}$$

- Where SE is standard error and depends on estimate of standard deviation
  - If standard deviation known: $SE = \frac{\sigma}{\sqrt{n}}$
  - If standard deviation not known: $SE = \frac{s}{\sqrt{n}}$
- Rejection region: depends on the distribution
  - If large sample or known variance $\text{test statistic} = Z_{test} \sim N(0, 1)$
    - Critical values come from standard normal distribution
  - If small sample with normal distribution $\text{test statistic} = T_{test} \sim t(n-1)$
    - Critical values come from student t with n-1 degrees of freedom

Single Sample - Mean - Two Sided Test

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu \neq \mu_0$

Rejection Region: For a test of significance level α, reject

- If large sample or known variance from normal

$$\frac{\bar{X} - \mu_0}{SE} < -z_{\frac{\alpha}{2}} \qquad or \qquad \frac{\bar{X} - \mu_0}{SE} > z_{\frac{\alpha}{2}}$$

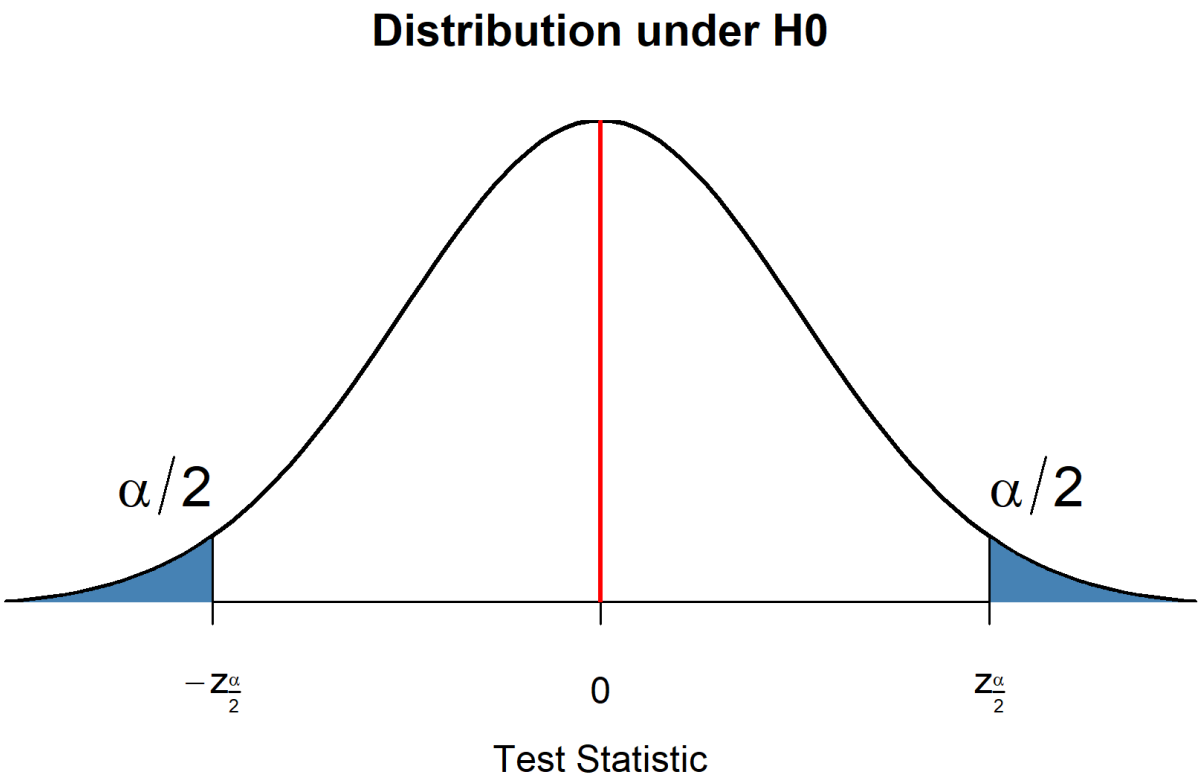
- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} < -t_{(n-1),\frac{\alpha}{2}} \qquad or \qquad \frac{\bar{X} - \mu_0}{SE} > t_{(n-1),\frac{\alpha}{2}}$$

Where

- $z_{\frac{\alpha}{2}}$ and $t_{(n-1),\frac{\alpha}{2}}$ are $(1 - \frac{\alpha}{2})$ quantiles of standard normal and t-students with n-1 degrees of freedom

Single Sample - Mean - Two Sided Test



Link between Tests and Confidence Intervals

Imagine a test for a parameter μ with hypothesis $H_0 = 4$ and $H_A \neq 4$.

- Suppose you draw a sample of 49 units with a mean $\bar{x} = 3$ and standard deviation is $s = 3$
- Our test statistic is $z_{test} = \frac{3-4}{3/7} = -2.34$
- Our critical values at $\alpha = 0.05$ are -1.96 and 1.96, hence we reject
- We can also calculate the 95% confidence interval for our sample mean:

$$\{3 - 1.96\frac{3}{\sqrt{49}}, 3 + 1.96\frac{3}{\sqrt{49}}\} = \{2.16, 3.84\}$$

- It does not contain the null hypothesis

Link between two sided Tests and Confidence Intervals

- More generally: the $1-\alpha$ confidence interval doesn't contain null = null would be rejected with a test of significance α

https://kzaremba.shinyapps.io/Hypothesis_Confidence/

- Mathematically:
  - Let μ_0 be the null hypothesis. We reject (in two sided test) the null at α if

$$\left|\frac{\bar{X}-\mu_0}{s/\sqrt{n}}\right| > z_{\frac{\alpha}{2}}$$

  - Which can be rewritten as:

$$\mu_0 < \bar{X} - z_{\frac{\alpha}{2}}\frac{s}{\sqrt{n}} \qquad or \qquad \mu_0 > \bar{X} + z_{\frac{\alpha}{2}}\frac{s}{\sqrt{n}}$$

> **LearnR:** CI and Hypothesis Test equivalence — verify that rejecting at `(\alpha)` matches the CI not containing `(\mu_0)`

> **LearnR:** One-Sample t-test — Test whether a Twitter campaign generates at least 200 likes per tweet

Single Sample - Mean - One sided - Case 1

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu < \mu_0$

- This includes any null hypothesis like $\mu \geq \mu_0$. Why?
- If you rejected μ_0 at α, you would reject for sure anything larger than μ_0

Rejection Region: For a test of significance level α, reject

- If large sample or known variance from normal

$$\frac{\bar{X} - \mu_0}{SE} < -z_\alpha$$

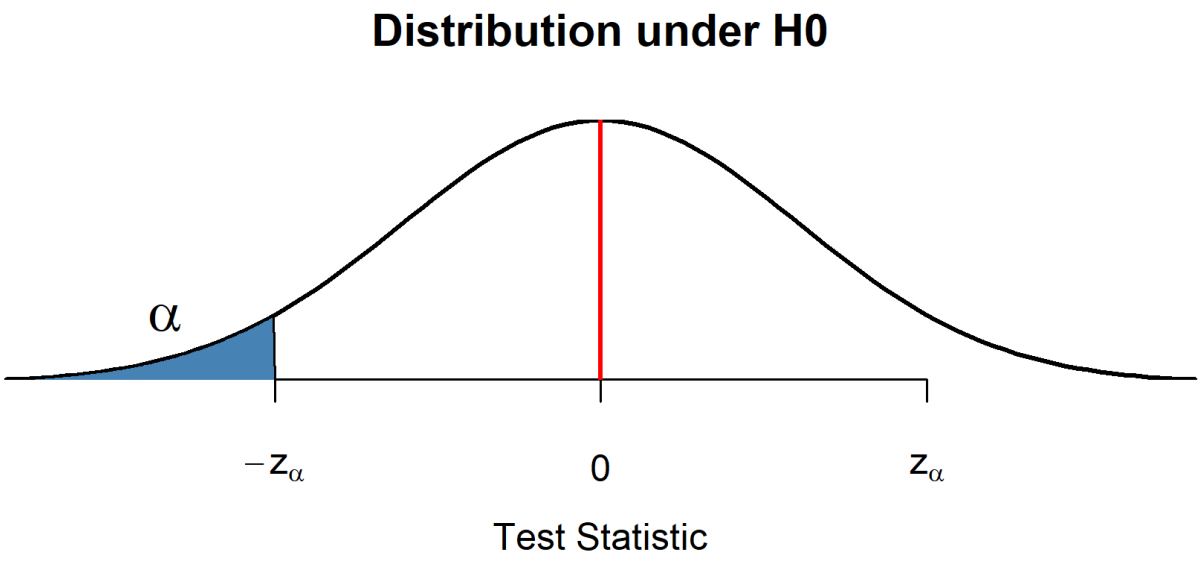
- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} < -t_{(n-1),\alpha}$$

Where

- z_α and $t_{(n-1),\alpha}$ are $(1 - \alpha)$ quantiles of standard normal and t-students with n-1 degrees of freedom

Single Sample - Mean - One sided - Case 1



- Intuition:
  - We never reject H_0 if $\bar{X} \geq \mu_0$.
  - We reject H_0 if $\bar{X}$ is sufficiently smaller than μ_0
  - If null is true ($\mu = \mu_0$) then we reject with probability

$$P(\frac{\bar{X} - \mu_0}{s/\sqrt{n}} < -z_\alpha) = P(Z < -z_\alpha) = \alpha$$

Single Sample - Mean - One sided - Case 2

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu > \mu_0$

- This includes any null hypothesis like $\mu \leq \mu_0$. Why?
- If you rejected μ_0 at α, you would reject for sure anything smaller than μ_0

Rejection Region: For a test of significance level α, reject

- If large sample or known variance from normal

$$\frac{\bar{X} - \mu_0}{SE} > z_\alpha$$

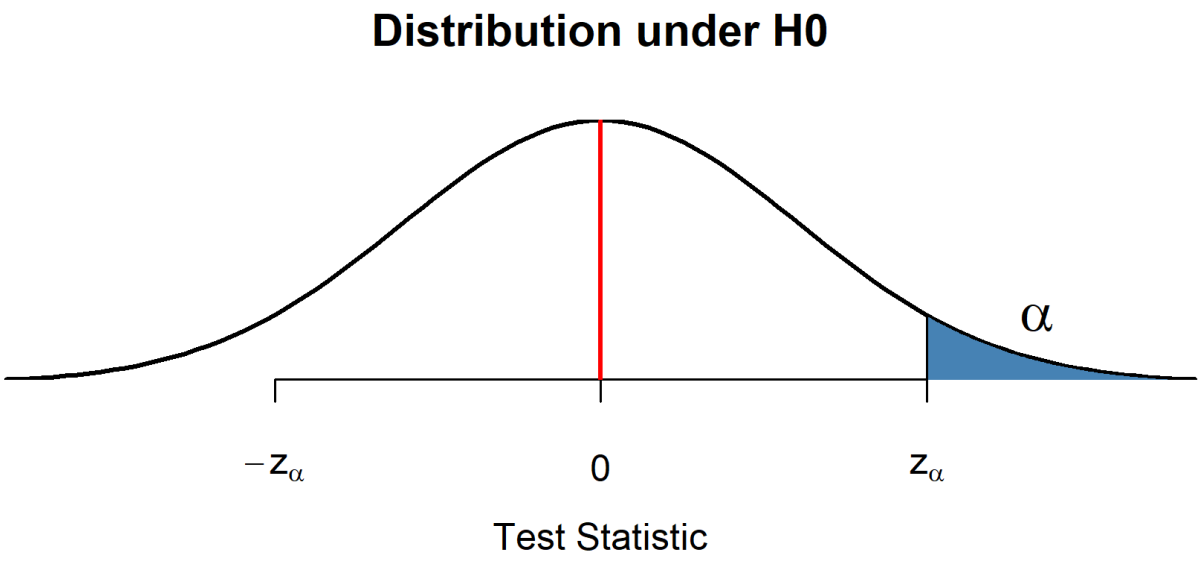
- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} > t_{(n-1),\alpha}$$

Where

- z_α and $t_{(n-1),\alpha}$ are $(1 - \alpha)$ quantiles of standard normal and t-students with n-1 degrees of freedom

Single Sample - Mean - One sided - Case 2



- Intuition:
  - We never reject H_0 if $\bar{X} \leq \mu_0$.
  - We reject H_0 if $\bar{X}$ is sufficiently larger than μ_0
  - If null is true ($\mu = \mu_0$) then we reject with probability

$$P(\frac{\bar{X} - \mu_0}{s/\sqrt{n}} > z_\alpha) = P(Z > z_\alpha) = \alpha$$

P-values

Definition: The p-value is the probability (between 0 and 1), under the null hypothesis, of obtaining a test statistic at least as extreme as the one calculated from the sample.

Depends on:

- Your hypothesis
- Calculated test statistic
- Suppose the test statistic under H_0 is $Z_{test} \sim N(0,1)$
- One sided tests:
  - For $H_A : \mu > \mu_0 \; p - value = P(Z_{Test} \geq (computed - test - statistic))$
  - For $H_A : \mu < \mu_0 \; p - value = P(Z_{Test} \leq (computed - test - statistic))$
- Two sided test:
  - For $H_A : \mu \neq \mu_0 \; p - value = 2P(Z_{Test} \geq |(computed - test - statistic)|)$

Example:

- We test for $H_0 : \mu = 5$ and $H_A : \mu > 5$
- Suppose in sample a of 36 we have $\bar{x} = 5.5$ and $s = 2$.
- Test statistic in this case is distributed normally

$$p - value = P(Z > \frac{5.5 - 5}{2/6}) = P(Z > 1.5) = 0.066$$

Intuitively: The probability that a statistic we calculated (or more extreme value) could arise just by chance if null is true

P-value in two-sided test

- In a two-sided test, the p-value accounts for both tails
- We double the one-tail probability: $p\text{-value} = 2P(Z \geq |z_{test}|)$

Interactive visualization: https://rpsychologist.com/pvalue/

> LearnR: P values — compute the p-value for a coin fairness test

P-values and Test Significance

- In our example we calculated the p-value of 0.066
- Would the test at $\alpha = 0.05$ reject the null?
- Any test with the significance level $\alpha > p - value$ would reject the null

Another way to think about p-values

- The P-value is the smallest significance level α at which the H_0 can be rejected

(3) The average grade among fourth graders on a math aptitude test in a certain state is 62.5 out of 100 points. (Assume 62.5 is the population average.) An educational company has developed an app for elementary-school math and would like to test whether the app improves average performance on the aptitude test among fourth graders. Let μ_X denote the population mean of test scores among students who use the app.

(a) To provide statistical evidence that the app *increases* the true average of test scores, what are the appropriate one-sided null hypothesis and alternative hypothesis?

(b) The company provides the app to 15 randomly selected students. The sample mean and sample standard deviation of test scores are 68.3 and 13.5, respectively. Under the assumption that the test scores are i.i.d. and normally distributed, would the t-test of the null hypothesis in (a) be rejected at the 5% level? at the 10% level?

(c) What is the p-value associated with the t-test in (b)?

Hypothesis Testing: Two samples

Two Samples and the Difference in Means

We have 2 iid samples independent of each other from two populations $X_1, X_2, .., X_n$ and $Y_1, Y_2, .., Y_m$
.

General idea:

- We test the difference between population means in two populations
  - Ex: $H_0 : \mu_X - \mu_Y = 100$ vs $H_A : \mu_X - \mu_Y \neq 100$
  - Ex: $H_0 : \mu_X - \mu_Y = 0$ vs $H_A : \mu_X - \mu_Y > 0$
  - Test for no difference:
    - Ex: $H_0 : \mu_X - \mu_Y = 0$ vs $H_0 : \mu_X - \mu_Y \neq 0$
- Test statistic: normalized difference in sample means

$$\text{test stat} = \frac{\bar{X} - \bar{Y} - \overbrace{(\mu_{X,0} - \mu_{Y,0})}^{\Delta_0}}{SE} = \frac{\bar{X} - \bar{Y} - \Delta_0}{SE}$$

Variance in two samples

- Where SE is standard error and depends on the estimate of standard deviation
  - If standard deviation known: $SE = \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$
- Why? Because $Var(\bar{X} - \bar{Y}) = Var(\bar{X}) + Var(\bar{Y}) = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$
- If standard deviation not known: $SE = \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$
- Rejection region: depends on the distribution
  - If large sample or known variance from normal $\text{test statistic} = Z_{test} \sim N(0, 1)$
    - Critical values come from standard normal distribution
  - If small sample with normal distribution $\text{test statistic} = T_{test} \sim t(v)$
    - Critical values come from student t with v degrees of freedom

Side note on the number of degrees of freedom for difference in means

Two ways to do it:

1. Proper, complex (and annoying):

$$v = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m}\right)^2}{\frac{(s_X^2/n)^2}{n-1} + \frac{(s_Y^2/m)^2}{m-1}}$$

Round down to nearest integer

1. Simple, approximate shortcut:

$$v = min(n-1, m-1)$$

Difference in means - Two Sided Test

Hypothesis: $H_0 : \mu_x - \mu_Y = \Delta_0$ and $H_A : \mu_x - \mu_Y \neq \Delta_0$

Rejection Region: For a test of significance level α, reject

- If large sample or known variance from normal

$$\frac{\bar{X} - \bar{Y} - \Delta_0}{SE} < -z_{\frac{\alpha}{2}} \qquad or \qquad \frac{\bar{X} - \bar{Y} - \Delta_0}{SE} > z_{\frac{\alpha}{2}}$$

- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - \Delta_0}{SE} < -t_{v,\frac{\alpha}{2}} \qquad or \qquad \frac{\bar{X} - \bar{Y} - \Delta_0}{SE} > t_{v,\frac{\alpha}{2}}$$

Where

- $z_{\frac{\alpha}{2}}$ and $t_{v,\frac{\alpha}{2}}$ are $(1 - \frac{\alpha}{2})$ quantiles of standard normal and t-students with v degrees of freedom

Difference in means - Two Sided Test

If CLT kicks in, the distributions are:

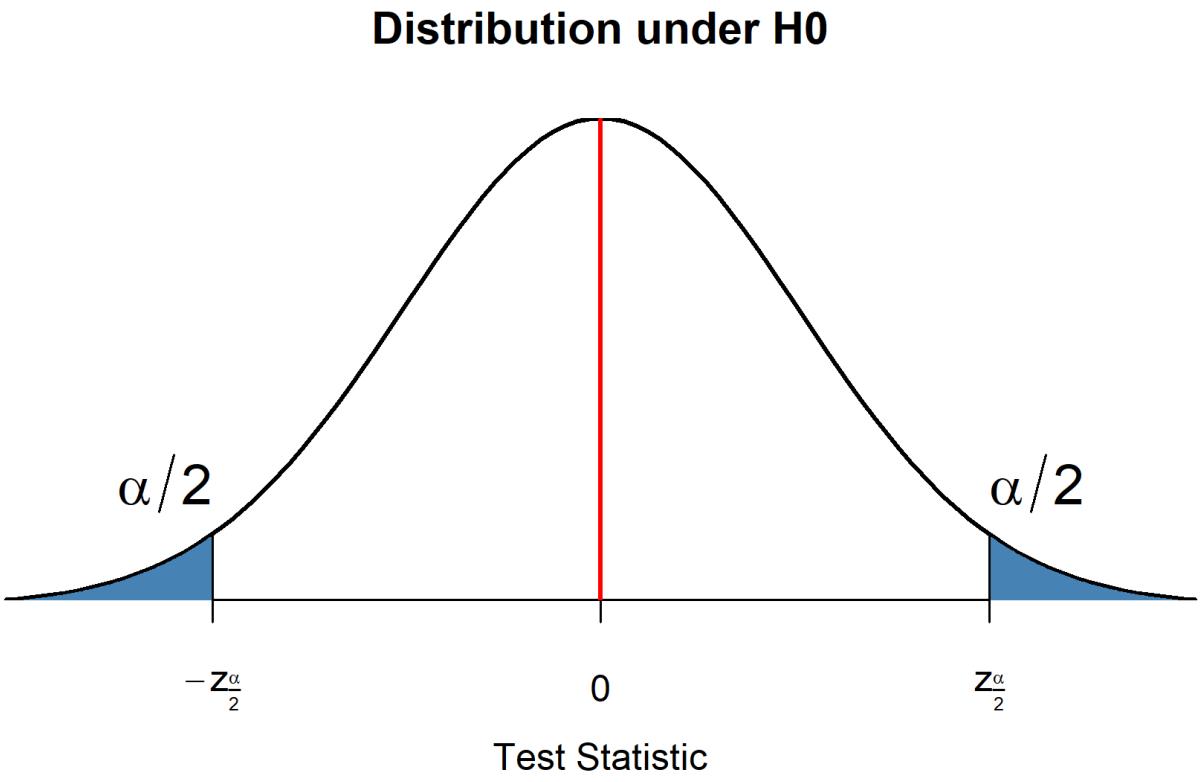
$$\bar{X} \sim N(\mu_X, \sigma_X/\sqrt{n}) \qquad and \qquad \bar{Y} \sim N(\mu_Y, \sigma_Y/\sqrt{m})$$

The difference of two normal (independent) distributions under:

$$\bar{X} - \bar{Y} \sim N(\mu_X - \mu_Y, \sqrt{\sigma_X^2/n + \sigma_Y^2/m})$$

So if null is true, then standardized statistic:

$$\frac{\bar{X} - \bar{Y} - (\mu_{X,0} - \mu_{Y,0})}{\sqrt{s_X^2/n + s_Y^2/m}} \sim N(0,1)$$


Distribution under H0
α/2
α/2
$-z_{\frac{\alpha}{2}}$
0
$z_{\frac{\alpha}{2}}$
Test Statistic

Confidence interval for a difference

Following this idea, we can construct a confidence interval for the difference:

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \bar{Y} - (\mu_{X,0} - \mu_{Y,0})}{\sqrt{s_X^2/n + s_Y^2/m}} < z_{\alpha/2}\right) = 1 - \alpha$$

$$CI_{1-\alpha} = \left(\bar{X} - \bar{Y} \pm z_{\alpha/2}\sqrt{s_X^2/n + s_Y^2/m}\right)$$

Two Samples - Mean - One sided - Case 1

Hypothesis: $H_0 : \mu_X - \mu_Y = \Delta_0$ and $H_A : \mu_X - \mu_Y < \Delta_0$

- This includes any null hypothesis like $\mu_X - \mu_Y \geq \Delta_0$. Why?
- If you rejected $\mu_X - \mu_Y = \Delta_0$ at α, you would reject for sure anything larger than Δ_0

Rejection Region: For a test of significance level α, reject

- If large sample or known variance from normal

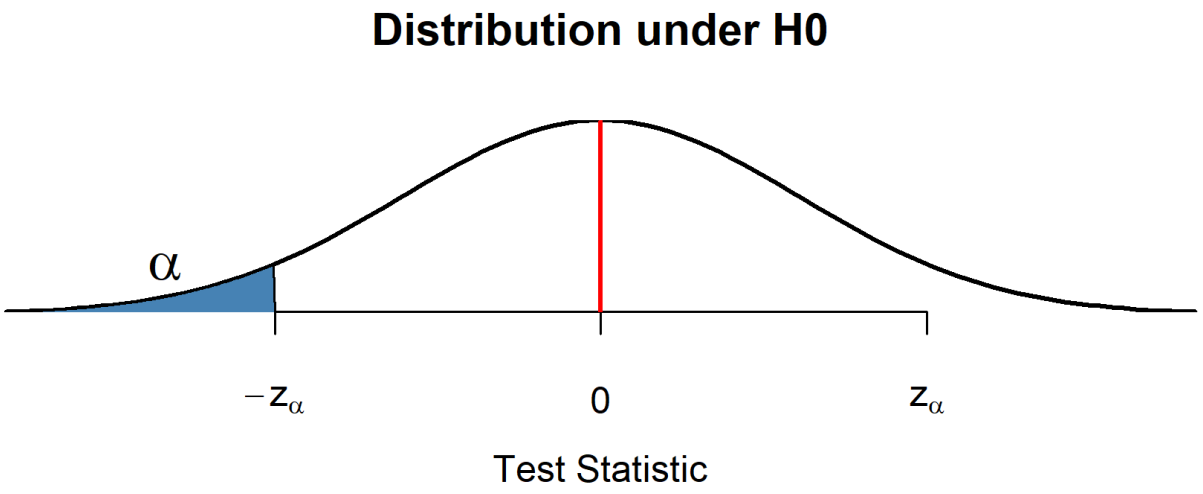
$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} < -z_\alpha$$

- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} < -t_{v,\alpha}$$

  Where
- $-z_\alpha$ and $-t_{v,\alpha}$ are α quantiles of standard normal and t-students with v degrees of freedom

Two Samples - Mean - One sided - Case 1



- Intuition:
  - We never reject H_0 if $\bar{X} - \bar{Y} \geq \Delta_0$.
  - We reject H_0 if $\bar{X} - \bar{Y}$ is sufficiently smaller than Δ_0
  - If null is true ($\mu_X - \mu_Y = \Delta_0$) then we reject with probability

$$P(\frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{s_X^2/n + s_Y^2/m}} < -z_\alpha) = P(Z < -z_\alpha) = \alpha$$

Two Samples - Mean - One sided - Case 2

Hypothesis: $H_0 : \mu_X - \mu_Y = \Delta_0$ and $H_A : \mu_X - \mu_Y > \Delta_0$

- This includes any null hypothesis like $\mu_X - \mu_Y \leq \Delta_0$. Why?
- If you rejected $\mu_X - \mu_Y = \Delta_0$ at α, you would reject for sure anything smaller than Δ_0

Rejection Region: For a test of significance level α, reject

- If large sample or known variance with normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} > z_\alpha$$

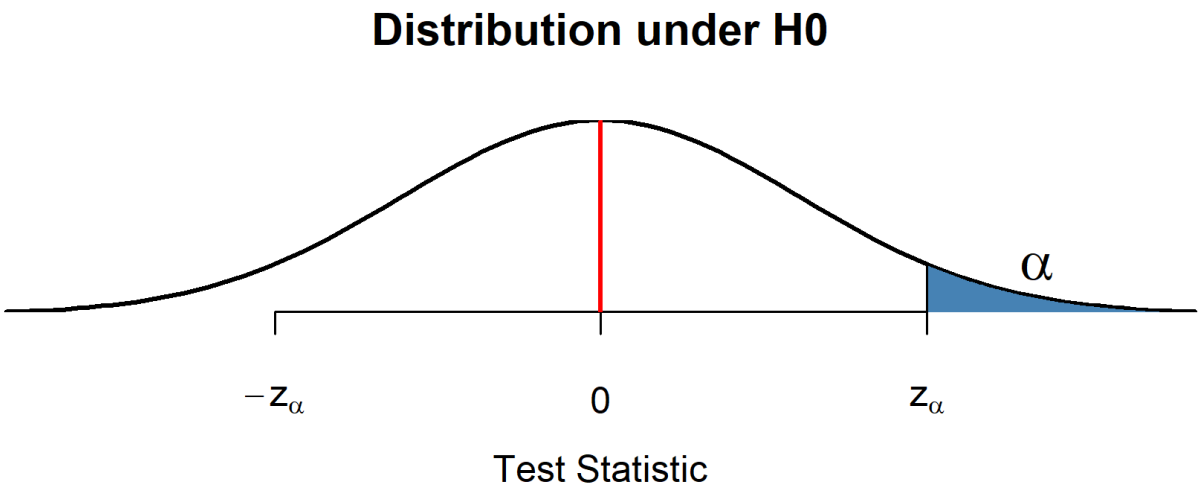
- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} > t_{v,\alpha}$$

Where

- z_α and $t_{v,\alpha}$ are $(1 - \alpha)$ quantiles of standard normal and t-students with v degrees of freedom

Two Samples - Mean - One sided - Case 2

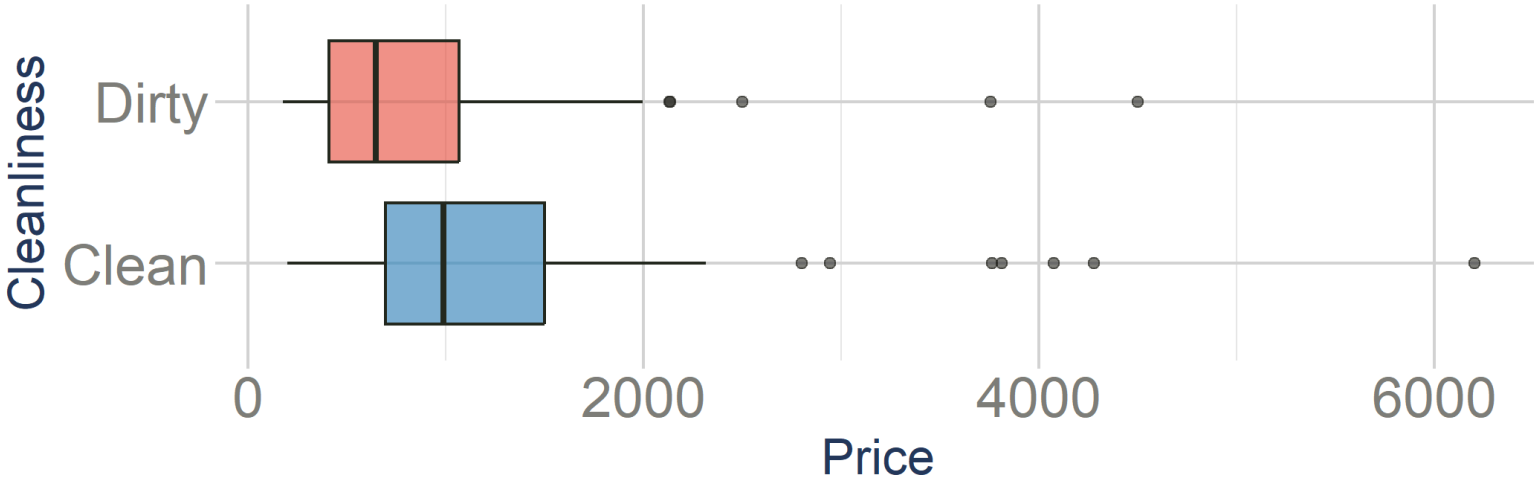


- Intuition:
  - We never reject H_0 if $\bar{X} - \bar{Y} \leq \Delta_0$.
  - We reject H_0 if $\bar{X} - \bar{Y}$ is sufficiently larger than Δ_0
  - If null is true ($\mu_X - \mu_Y = \Delta_0$) then we reject with probability

$$P(\frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{s_X^2/n + s_Y^2/m}} > z_\alpha) = P(Z > z_\alpha) = \alpha$$

Airbnb: Clean vs Dirty Prices

Sample	n	Mean	SD
Clean	100	1245	962
Dirty	100	869	693



- State the null and alternative hypothesis
- Calculate the value of the test statistic
- Calculate the p-value
- At which α do we reject the null?
- Calculate 95% confidence interval for the difference

LearnR: Two Sample Difference — Calculate test statistic and p-value for this comparison

LearnR: Using `t.test()` in R — learn the built-in function for one-sample, two-sample, and paired tests

Exercise: McDonald's vs Burger King

We suspect Burger King might have larger caloric content for kids' meals. Let's test it.

Sample	n	Sample Mean	Sample Standard Deviation
McDonald	9	475	230
Burger King	16	520	180

- State the null and alternative hypothesis
- Calculate the value of the test statistic
- Determine the appropriate number of degrees of freedom
- Calculate the p-value
- At which α do we reject the null?

> LearnR: Small-sample t-test — work through this McDonald's vs Burger King comparison with exact df

A/B Testing: The Business Question

- Suppose Airbnb wants to know:
  Do professional photos increase host revenue?
- Historical data: Listings with better photos show higher revenues.
- But… Correlation ≠ Causation.
  - Better photos may be linked to other factors:
  - More organized hosts
  - Higher quality listings
  - Better customer service
- Revenues may reflect these differences, not the photos themselves.
- A simple comparison only tells us about association, not causality.

The Business Decision

- Should Airbnb invest in professional photography for hosts?
- If photos aren't the true driver → investment wastes money.

We need a method to isolate the effect of photos.

Enter A/B Testing

Definition:

- A method to measure *causal* effects.
- Widely used in modern business, especially tech.

Examples:

- Netflix: Which thumbnail gets more clicks?
- Amazon: Does a new checkout button increase sales?
- Instagram: Do longer videos increase engagement?

The Setup

We divide hosts into two groups:

1. Control group:
    - No intervention (existing photos).
2. Treatment group:
    - Airbnb provides professional photography.

The KEY:

- Groups are assigned randomly.

  Why Randomization Matters

- Without randomization: groups differ in many ways.
- With randomization: groups are (on average) identical, except for photos.

This is what gives us causal inference!

Classroom Exercise: Randomization Check

- Take a sample of Airbnb listings
- Randomly assign to two groups
- Compare characteristics (beds, location)

They should look very similar!

> LearnR: AB Testing Randomization — sample listings, randomize, and check balance

Why? We are taking two samples from the same population.

- All properties and expectations will be the same for both samples

Measuring the Effect

- After randomization: the only difference is the photos

- Provide photography to treatment group

- Wait to see potential effects
- Collect outcome data (revenue, reviews, bookings)
- Run a simple two-sample test between treatment and control
- If the treatment group earns more revenue, we know it was caused by the professional photos

This is the causal effect.

- A/B testing is used extensively in modern business, especially in tech
- We have all been subjects of these experiments, often without knowing

Hypothesis Testing: Paired Data

Paired Data

- Sometimes we have only one set of individuals or objects, but two (or more) observations per each.
- Example: We look at employees productivity. For each employee we have two observations: their productivity when working from home and their productivity when working from the office.
- Example: We look at sales at Walmart and Soriana in each neighborhood. We have two observations per neighborhood: sales at local Walmart and sales at local Soriana

Paired Data

We have a set of independent pairs of data $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$ Observation within the pair are not necessarily independent!

- Some productive workers are productive both in the office and at home, some unproductive are unproductive both in the office and at home.
- We are interested in the difference between X and Y. We observe that difference for each individual in the sample: $X_i - Y_i = d_i$
- We want to look at the population value of that difference: $\mu_X - \mu_Y = \Delta$
  - We can check if the means are the same: $\Delta = 0$
    - No difference in productivity between work at home and at the office
  - Or mean X is bigger than mean Y: $\Delta > 0$

Paired Data

Null hypothesis: $H_0 : \Delta = \Delta_0$

Test Statistic is:

$$\text{test statistic} = \frac{\overbrace{\bar{X} - \bar{Y}}^{\bar{d}} - \Delta_0}{SE} = \frac{\bar{d} - \Delta_0}{SE}$$

- Note on the Standard Error - it's not the same formula as for the two independent samples: $Var(X_i - Y_i) \neq Var(X_i) + Var(Y_i)$
- We just calculate it from the differences.
  - If we know the variance, $SE = \frac{\sqrt{var(d_i)}}{\sqrt{n}} = \frac{\sigma_d}{\sqrt{n}}$
  - If we don't: $SE = \frac{s_d}{\sqrt{n}}$

Paired Data

Rejection Region: For a test of significance level α:

- If large sample or known variance with normal distribution:
- for $H_A : \bar{d} \neq \Delta_0$, reject if:

$$\frac{\bar{d} - \Delta_0}{SE} < -z_{\frac{\alpha}{2}} \qquad or \qquad \frac{\bar{d} - \Delta_0}{SE} > z_{\frac{\alpha}{2}}$$

- for $H_A : \bar{d} > \Delta_0$, reject if:

$$\frac{\bar{d} - \Delta_0}{SE} > z_\alpha$$

- for $H_A : \bar{d} < \Delta_0$, reject if:

$$\frac{\bar{d} - \Delta_0}{SE} < -z_\alpha$$

- If small sample from normal, replace critical values with t-distribution with n-1 degrees of freedom

Paired or Unpaired Data?

1. We would like to know if Intel's stock and Southwest Airlines' stock have similar rates of return. To find out, we take a random sample of 50 days, and record Intel's and Southwest's stock on those same days

- Paired

1. We randomly sample 50 items from Target stores and note the price for each. Then we visit Walmart and collect the price for each of those same 50 items.

- Paired

1. A school board would like to determine whether there is a difference in average SAT scores for students at one high school versus another high school in the district. To check, they take a simple random sample of 100 students from each high school

- Not paired

> LearnR: Paired or Unpaired? — practice classifying four more business scenarios

Exercise: Does IQ Decrease with Age?

A psychologist thinks that age influences IQ. They take a random sample of 100 people aged 40. For each person we know their IQ at age 16 and now. On average, IQ at young age was 8 points higher than at age 40. The standard deviation of that difference was 7 points. Using $\alpha = 0.01$, test the hypothesis that IQ decreases with age.

> LearnR: Paired Data test for means — work through this IQ example in R

Hypothesis Testing for Correlation

Example: are cleanliness score (C) and price (P) correlated in the population?

Assume you have normally distributed variables in independent pairs. $\{(C_1, P_1), (C_2, P_2), \dots\}$

Null hypothesis: $H_0 : \rho(C, P) = \rho_0$

Test statistic and its distribution under the null:

$$T_{test} = \frac{\hat{\rho}(C, P) - \rho_0}{\sqrt{\frac{1-\hat{\rho}(C,P)^2}{n-2}}} \sim t_{n-2}$$

where $\hat{\rho}(C, P)$ is sample correlation coefficient

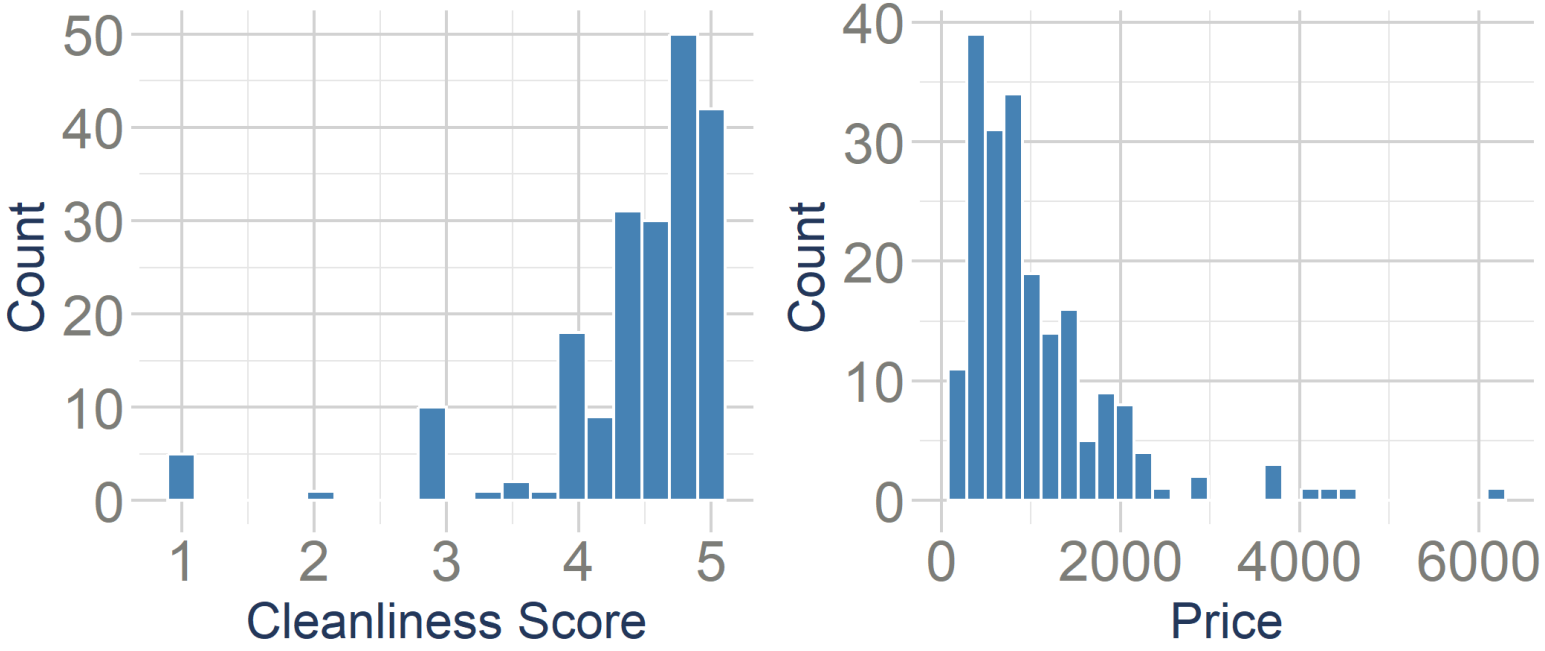
Alternative hypothesis and their rejection regions:

- $H_A : \rho(C, P) \neq \rho_0$ - reject H_0 if $t_{test} > t_{\frac{\alpha}{2}, n-2}$ or $t_{test} < -t_{\frac{\alpha}{2}, n-2}$
- $H_A : \rho(C, P) > \rho_0$ - reject H_0 if $t_{test} > t_{\alpha, n-2}$
- $H_A : \rho(C, P) < \rho_0$ - reject H_0 if $t_{test} < -t_{\alpha, n-2}$

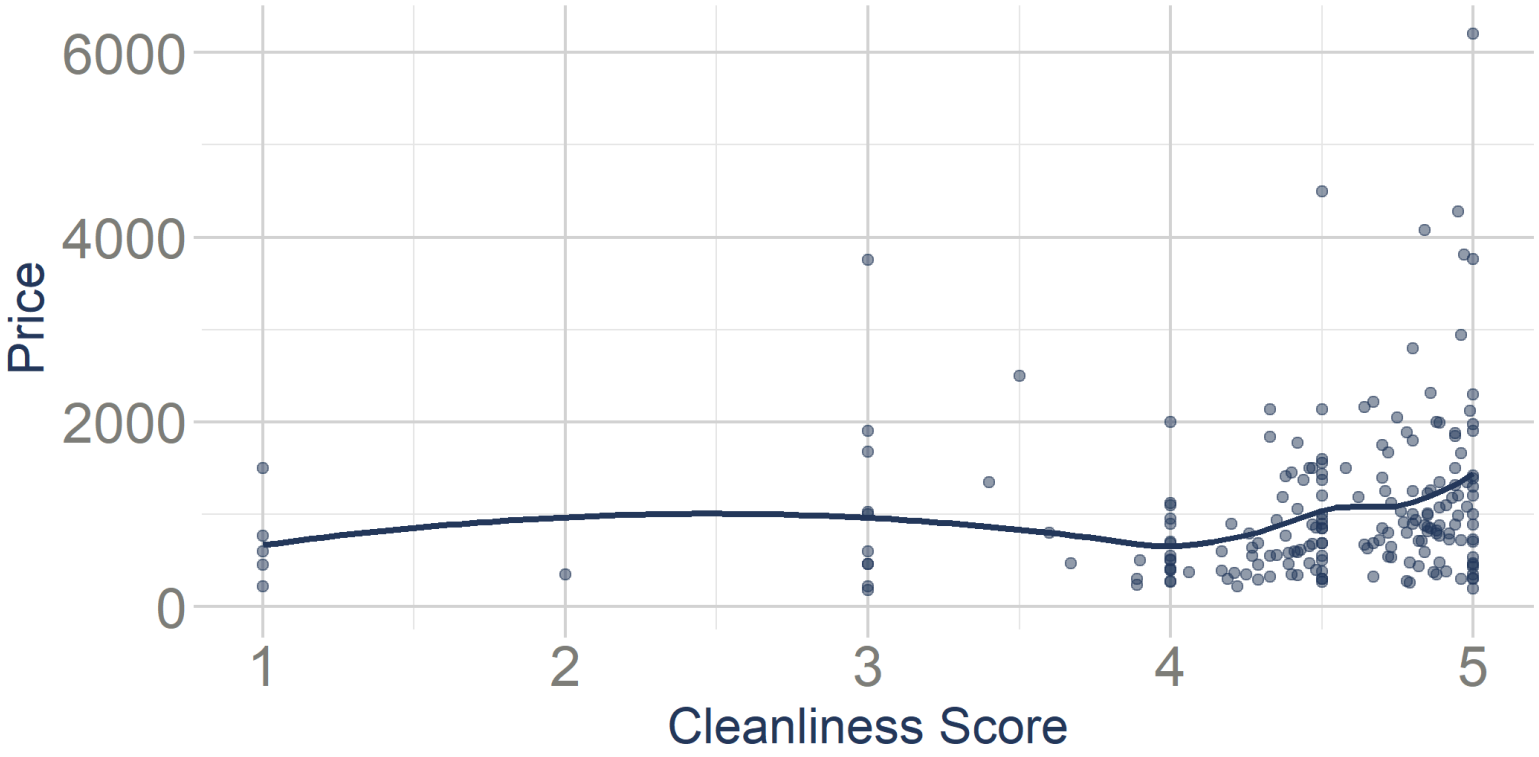
Correlation test

- Suppose we test $H_0 : \rho(C, P) = 0$ vs $\rho(C, P) > 0$ in our Airbnb Data (clean score vs price)
- We assume that:
  - The variables are normally distributed
  - Pairs are independent
  - Relationship would be linear
- Sample correlation is: 0.159, we have $n = 200$
- So $t_{test} = \frac{\hat{\rho}(C,P)\sqrt{(n-2)}}{\sqrt{1-\hat{\rho}(C,P)^2}} = \frac{0.159\sqrt{198}}{\sqrt{1-0.025}} = 2.27$
- Since $n = 200$, t-student is identical to standard normal. We can either
  - Compare calculated statistic to critical regions at various significance levels
  - Compute its p-value
- $p - value = P(Z > t_{test}) = P(Z > 2.27) = 0.0126$
  - Test at $\alpha = 0.05$ would reject H_0

Correlation Test — Distributions



Correlation Test — Scatter Plot



Exercise: Correlation Test

Suppose you have a random sample of 27 units (from a bivariate normal). X measures client's age and Y measures their spending. You calculated the correlation coefficient $\hat{\rho} = -0.45$. Can you reject the null $\rho = 0$ in favor of the alternative $\rho \neq 0$ at the 5% significance level?

> LearnR: Paired Data Test for Correlation — compute the t-statistic for this exercise